

IAL MECHANICS 1 · RAPID REVISION SHEET

Pearson Edexcel WME01/01 · Distilled from 19 papers and mark schemes, January 2020 – Summer 2026

1 FIVE RULES THAT DECIDE YOUR GRADE

FINAL-ANSWER RULES

- After using $g = 9.8$: answer to **2 or 3 s.f.** as a decimal. $\frac{5}{8}, \frac{32}{125}, "2g - 3"$ were all A0.
- No** g in any momentum or impulse equation (M0).
- Write every equation **in full, unsimplified** first: that line is the A1.
- "Show that": end on the **exact printed form** ($\frac{4}{5}g$, not $0.8g$).
- Speeds and impulse magnitudes positive; impulse in Ns; bearings in 3 figures.

M1 needs *the right number of terms, dimensionally correct*; sign errors and sin/cos swaps are condoned for M but cost the A.

2 KINEMATICS

SUVAT (CONSTANT a ONLY)

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad s = vt - \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as \quad s = \frac{1}{2}(u + v)t$$

- Vertical:** up $+$, $a = -g$. Top: $t = U/g$, $H = U^2/2g$. Landing h below start: $s = -h$.
- New stage** whenever a changes; final $v \rightarrow$ next u . One suvat for the whole motion = M0.
- Time above a level** = difference of the two roots of the quadratic.
- n th second:** $s(n) - s(n-1)$. **Total distance** = up + down.
- Catch-up:** same clock t ; late starter moves for $(t - t_0)$; equate distances.
- Graphs:** $v-t$ gradient = a , area = displacement; $a-t$ area = Δv ; $s-t$ gradient = v .
- Sketches:** vertical motion speed-time is a V with vertex *on* the t -axis; velocity-time is one straight line crossing the axis. Never a solid vertical line. Label start, end, turning values.

3 FORCES AND FRICTION

- $F \leq \mu R$; "on the point of" / "limiting" $\Rightarrow F = \mu R$.
- Friction opposes motion or *impending* motion. Greatest P : friction down slope. Least P : friction up slope.
- Find R first** (resolve $\perp$ surface, all forces):
 - pull at θ up: $R = mg - P \sin \theta$; push at θ down: $R = mg + P \sin \theta$
 - slope: $R = mg \cos \alpha$; horizontal X on slope: $R = mg \cos \alpha + X \sin \alpha$

- Recompute R whenever a force changes (old $R = M0$).
- Stays at rest on slope alone $\iff \mu \geq \tan \alpha$. Conclude *in words*.

EXACT TRIG

$\tan \alpha$	$\sin \alpha$	$\cos \alpha$	Note
$\frac{3}{4}$	$\frac{3}{5}$	$\frac{4}{5}$	most common
$\frac{5}{12}$	$\frac{5}{13}$	$\frac{12}{13}$	
$\frac{7}{24}$	$\frac{7}{25}$	$\frac{24}{25}$	Oct 2023

Use fractions, never a rounded angle.

4 ACCELERATIONS TO KNOW COLD

Situation	$ a $
Sliding on rough horizontal	μg
Down rough slope	$g(\sin \alpha - \mu \cos \alpha)$
Up rough slope (decelerating)	$g(\sin \alpha + \mu \cos \alpha)$
Down smooth slope	$g \sin \alpha$
Two hanging masses $M > m$	$\frac{(M - m)g}{M + m}$
Hanging M pulls m on smooth table	$\frac{Mg}{M + m}$

5 CONNECTED PARTICLES

- One $F = ma$ per particle**, each with its own mass; same T , same $|a|$. Whole-system equation cannot give T .
- Hanging particle accelerating: $T \neq mg$ (using $T = mg$ loses marks).
- Lift:** person $R - mg = ma$; whole lift uses total mass and cable T .
- Towing:** whole system first, then one vehicle. A light-rod coupling may be a *thrust*.

FORCE ON PULLEY

$2T \cos \frac{\phi}{2}$, along the bisector, ϕ = angle between strings.

Both vertical: $2T$. Table edge: $\sqrt{2}T$ at 45° .

Slope α + hanging: $\phi = 90^\circ - \alpha$.

SECOND PHASE

Ground hit or string breaks $\Rightarrow$ string slack. Other particle keeps its speed, gets a **new** a ($-\mu g$, $-g(\sin \alpha + \mu \cos \alpha)$ or $-g$). "Does it reach the pulley?": distance moved + extra slide vs available length, comparison written.

6 MOMENTUM AND IMPULSE

- CLM: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ with signs.
- Impulse = $m(v - u)$; on $P = -$ on Q . Either particle gives it.
- Coalesce / string goes taut: common velocity.
- Bounce: impulse = $m(v + u)$ using speeds; rebound speed from height via $v^2 = 2gh$.
- Unknown direction: assume +; negative $\Rightarrow$ reversed (state it).
- Only a **speed** given after $\Rightarrow$ **two cases** ("two values of k ").
- "Direction reversed" gives an inequality ($v > 0$).

7 MOMENTS

- Equilibrium: $\sum F_{\uparrow} = 0$ and $\sum M = 0$ about *any* point.
- Take moments about the force you **do not** want.
- **Tilting about C** $\Rightarrow$ **other reaction** = 0 (write it).
- Uniform: weight at midpoint. Non-uniform: at d from a named end.
- Two scenarios $\rightarrow$ two equations $\rightarrow$ eliminate (e.g. $M(d - 2.5) = 40$, $M(6 - d) = 72 \Rightarrow M = 32$).
- Range of a mass: every reaction ≥ 0 ; tensions $\leq$ breaking value.
- Never assume the two reactions are equal.

8 VECTORS

$$|ai + bj| = \sqrt{a^2 + b^2}; \quad \sum \mathbf{F} = m\mathbf{a}; \quad \mathbf{r} = \mathbf{r}_0 + \mathbf{v}t;$$

$$\mathbf{v} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}; \quad \mathbf{v} = \mathbf{u} + \mathbf{a}t; \quad \overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

Wording	Translate to
Parallel to $p\mathbf{i} + q\mathbf{j}$	components in ratio $p : q$
Due N / due E	$\mathbf{i} / \mathbf{j}$ part of $\overrightarrow{AB}$ is 0
NE / NW of A	$\overrightarrow{AB} = k(\mathbf{i} + \mathbf{j}) / k(-\mathbf{i} + \mathbf{j}), k > 0$
Collide	$\mathbf{r}_A = \mathbf{r}_B$; both parts give same t
Never collide	times zeroing each part differ
Closest	minimise $ \overrightarrow{AB} ^2$ (complete square)
Within d	$ \overrightarrow{AB} ^2 \leq d^2$; roots' difference
Same speed	$ \mathbf{v}_A ^2 = \mathbf{v}_B ^2$
Clock / units	36 min = 0.6 h; $\text{km h}^{-1} \div 3.6$
Bearing of B from A	sketch $\overrightarrow{AB}$; clockwise from $\mathbf{j}$ (north); 3 figures

9 ONE-MARK MODELLING STATEMENTS

Smooth pulley	tension same on both sides of the pulley
Light string	tension same throughout each straight section
Inextensible	both particles have the same acceleration
Rod	beam is rigid / does not bend
Uniform	weight acts at the midpoint
Particle	mass concentrated at a single point
More realistic	rough pulley, string with mass, size of object

One statement only: any incorrect extra = B0.

10 TOP TRAPS

1. Assuming $u = 0$ when it is not stated.
2. Mixing up $\overrightarrow{AB}$ and $\overrightarrow{BA}$.
3. Old R reused after a force changes.
4. Tonnes not converted ($15 \text{ t} = 15\,000 \text{ kg}$).
5. Using the whole-system equation to find T .
6. An inequality that never becomes an equation.
7. Quadrant error in a bearing (no sketch).
8. Forgetting the second root / second case.
9. Using a rounded angle instead of exact trig.
10. $g = 9.81$ (penalised once per question).

11 IN THE EXAM ROOM

EVERY QUESTION

Diagram $\rightarrow$ positive direction $\rightarrow$ name the step ("Res $\perp$ ", " $M(C)$ ", "CLM", "For B ") $\rightarrow$ full equation $\rightarrow$ solve $\rightarrow$ 3 s.f. and units. Budget ≈ 1.2 min per mark; if stuck, write the equations anyway: method marks do not need the answer.